



USER GUIDE

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INTRODUCTION

X-Min Learn is an open source, standalone software designed for the automatic identification of mineral phases from X-ray map data. It offers a comprehensive and interactive environment for analyzing your data using machine learning classifiers and image analysis algorithms.

The software also includes tools for developing custom machine learning models with a fully no-code approach. You can generate ground truth datasets from your selected rock samples, train models tailored to your specific research needs, evaluate their performance using statistical metrics and interactive graphics, and accelerate future classifications through your personally trained classifiers.

This user guide provides a thorough description of all X-Min Learn functionalities, with a particular focus on its analysis tools.

INTERFACE OVERVIEW

The interface of X-Min Learn consists of four distinct UI elements (see Figure 1):

- *Central widget*, where main tools are docked for quick and compact interaction.
- *Panes*, that provide extra information linked to data displayed in the central widget.
- *Toolbars*, which grant quick access to the main app functionalities.
- *Menu bar*, where all features can be accessed.

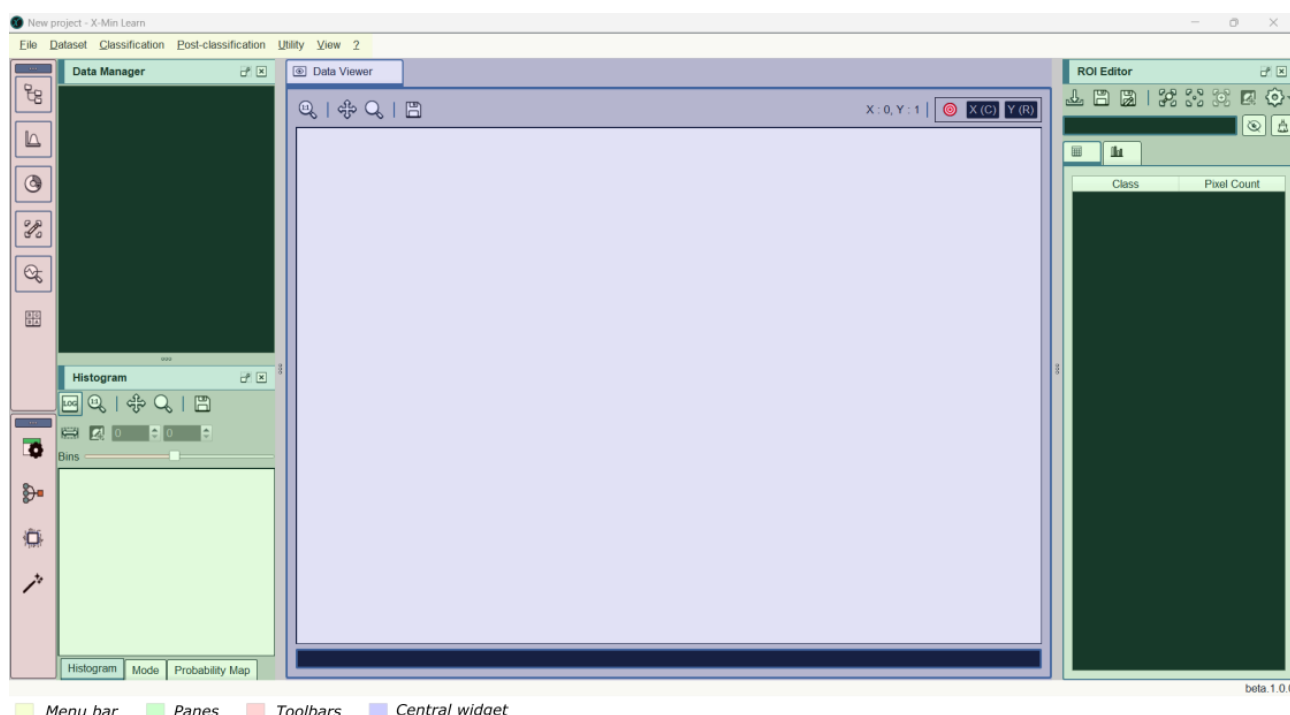


Figure 1 - Main Window interface.

The main functionalities of X-Min Learn are provided by the Analysis Tools, allowing for basic operations like data visualization (Data Viewer tool), classification (Mineral Classifier tool) and post-processing (Phase Refiner tool), and more advanced ones like custom machine learning models development (Dataset Builder and Model Learner).



Analysis tools are docked by default in the *central widget* (Figure 1) in a compact tabbed layout. Docked tools can be undocked, becoming standalone windows, by double clicking on their tab. Undocked tools can be docked back by dragging them on the *central widget* area. Tools can be opened from the *menu bar* or, more conveniently, from the *tools toolbar* (Figure 1).

The Data Viewer  is a special type of tool; it is always opened and cannot be closed nor undocked. Here your image data will be displayed. Simple navigation operations (e.g., zoom, pan, save image) are provided in its dedicated toolbar. Classic *mouse wheel interactions* are also available: scrolling for zooming (precise zoom if *Ctrl* key is pressed) and click-drag for panning. Use the *zoom to pixel* widget  to zoom to specific pixel coordinates.

Panes can be toggled from the dedicated *panes toolbar* (Figure 1). Panes position in the main window can be customized through drag & drop interaction. They can be placed on either side of the *central widget* and be organized in tabs by stacking them on top of each other. Like tools, panes can be undocked and docked back to the main window.

IMPORTING DATA

X-Min Learn schematically organizes mineral data into *samples*. A sample consists of:

- *Input maps*, gathering image data collected via lab analysis (X-ray maps, BSE maps, etc.).
- *Mineral maps*, that include image data derived from the analysis of input maps.
- *Masks*, for visually clipping input and mineral maps without changing their data.

Data can be imported from the *Menu bar* in File > Import...> *Input Maps* or *Mineral Maps* or *Masks*; this will automatically create a new sample in the Data Manager pane, populating the corresponding data type entry. Samples and their data can be directly managed from the Data Manager; each data type is characterized by a specific file format:

- *Input maps* must have the ASCII (*txt*) or compressed ASCII format (*gz*). The file must contain an array of pixel values separated by a space character.
- *Mineral maps* must have the *mmp* file extension or the same ASCII formats (*txt* and *gz*) specified for input maps. These last formats are available to provide cross compatibility with other software, but will be converted on the fly to *mmp*.
- *Masks* must have the *msk* format, and can be generated in different ways (see Extracting Masks).

Standard image formats (*bmp*, *jpg*, *png*, *tiff*) can be converted to valid map formats through dedicated dialogs (see Image Conversion).

PROJECTS

Your workspace can be saved in projects files with a dedicated file extension (*xmj*). Click on the File menu in the *Menu bar* to create, open or save projects.

Warning!

At the moment, data processed in other tools than the Data Viewer is NOT saved, so keep that in mind when closing your current project. It is planned to add this functionality in future releases.



PANES

Panes generally hold information that are related to the data currently displayed in the *central widget*, especially in the Data Viewer, and can be toggled from the dedicated *panes toolbar*. They are:

- [] Data Manager, that allows loading and managing samples data.
- [] Histogram, displaying histograms of input maps.
- [] Mode, which shows the percentage amount of each mineral class in mineral maps.
- [] ROI Editor, that allows tracing Regions of Interest (ROIs) on maps.
- [] Probability Map, displaying the classification confidence of mineral maps' pixels.
- [] RGBA Map, which allows building composite RGBA images by combining input maps.

Data Manager

The Data Manager is the most important pane, allowing the management of samples' data. Samples are organized in a tree view, following the scheme described in [Importing Data](#); each sample groups three main data types: input maps, mineral maps and masks. To add a new empty sample, right click on an empty space in the Data Manager and click on *New sample*. Then, right-click on the sample and select *Import... > Input Maps* or *Mineral Maps* or *Masks*, to add data. Alternatively, directly right-click on the specific data type sub-group and click on *Import*.

When left clicking on any map, it will be displayed in the Data Viewer. When data entries are edited, an *edit* icon [] is displayed next to them. Right-click the entry and select *Save* or *Save as...* to save edits on file, or select *Refresh source* to cancel all edits. When data file paths are altered (files are moved, renamed or deleted) a *warning* icon [] will appear next to the related data entries. Right-click the entry and select *Fix data source* to attempt fixing data path links. Unique options are available when right-clicking on the different data types:

- Input maps options:
 - *Invert*, to set the higher pixel values as the lower and vice versa.
 - *Set as RGBA channel...*, to set map as a channel of a composite [RGBA Map](#).
- Mineral maps options:
 - *Palette*, to randomize, copy or paste the palette (i.e., display color of mineral phases).
 - *Export map*, to export the mineral map as a numerical array in ASCII format (*txt*).
- Masks options:
 - *Invert*, to invert the mask (boolean inversion).
 - *Merge*, to merge two or more selected masks into a new one.

Histogram

When an input map is displayed in the Data Viewer, its histogram can be checked in the Histogram Pane. The plot can be adjusted according to your preferences (logarithmic scale, zoom, pan etc.) using the navigation toolbar. The *bins slider* can be used to adjust the number of bins.

For an advanced interaction with the plot, you can toggle the *scaler* [], which allows you to select a specific range of values from the histogram. The dedicated spin-boxes can be used to precisely



set upper and lower ranges. The selection will be reflected on the displayed map, changing the scale of its color bar accordingly. Use this tool to highlight hidden patterns in your input data, that may be flattened by the full range of pixel values (see Figure 2).

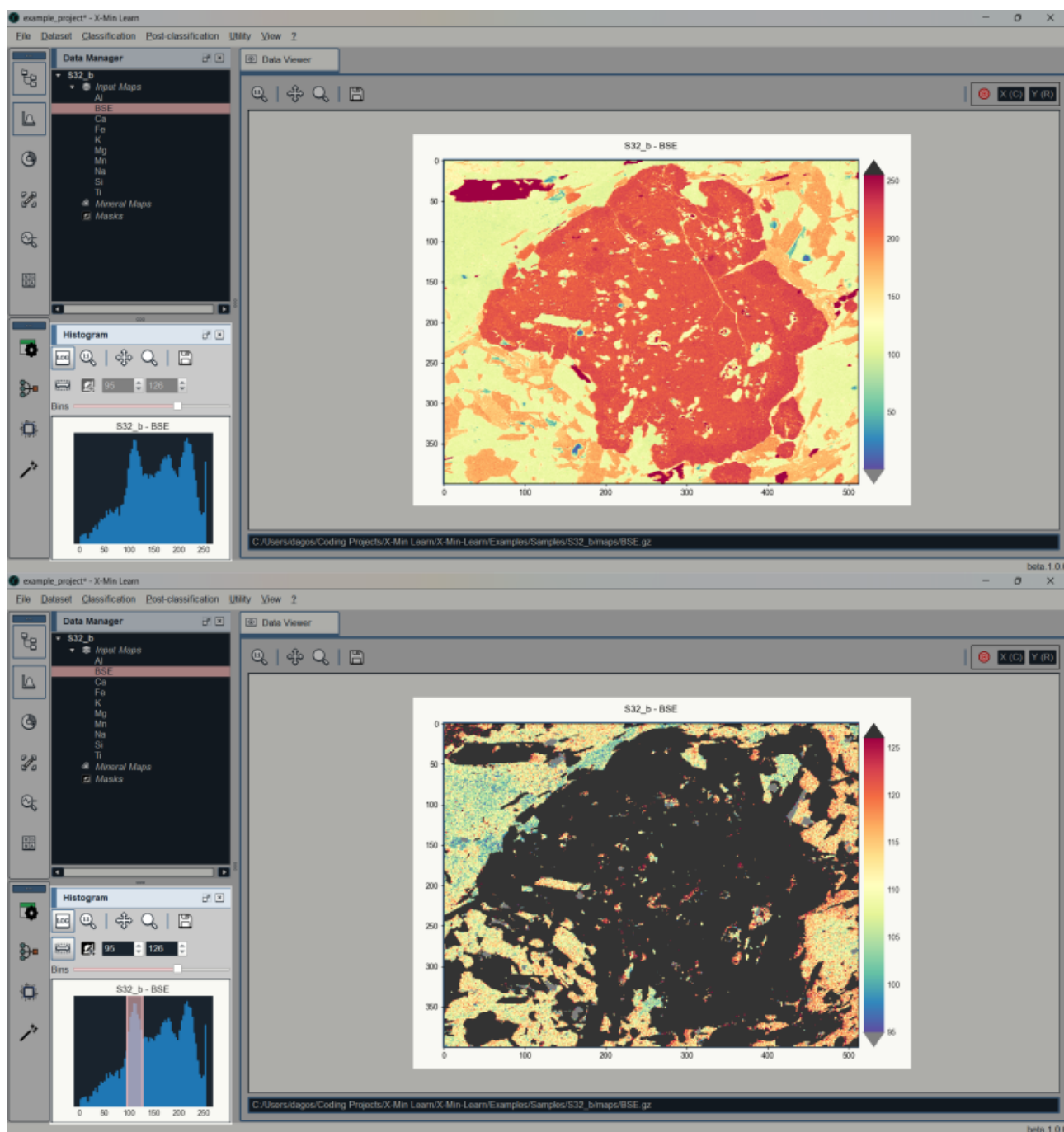


Figure 2 – Input map before and after the application of the histogram scaler.

Mode

Mineral maps are displayed in the Data Viewer in false colors, each representing a specific mineral class. The Mode pane allows you access extra information and customize the appearance of mineral maps. It consists of two tabs:

-  The *interactive legend*, which allows you to change names and display colors of mineral classes, merge two or more classes together and highlight one class against the others.



- [📊] The *bar plot*, that visually displays the amount of each mineral class.

ROI Editor

A Region of Interest (ROI) can be drawn on the displayed map using the ROI Editor Pane, by enabling the *draw ROI* button [📐] in the pane's toolbar. ROIs have two main uses:

- Highlighting in the Histogram the range of pixels within the ROI.
- Populate a ROI map. A ROI map is a special map-like object that groups multiple user-labeled ROIs. It is required by ROI-based algorithms to classify mineral data (see Mineral Classifier). Use the buttons in the *ROI* toolbar to save/load a ROI map. To add the currently drawn ROI to a ROI map, click the *add ROI* button [📐] in the toolbar. A popup will appear, allowing you to assign a label (mineral class) to the ROI. Added ROIs are listed in the *ROI table* tab [📄]; the number of labeled pixels per mineral class is displayed in the *bar plot* tab [📊].

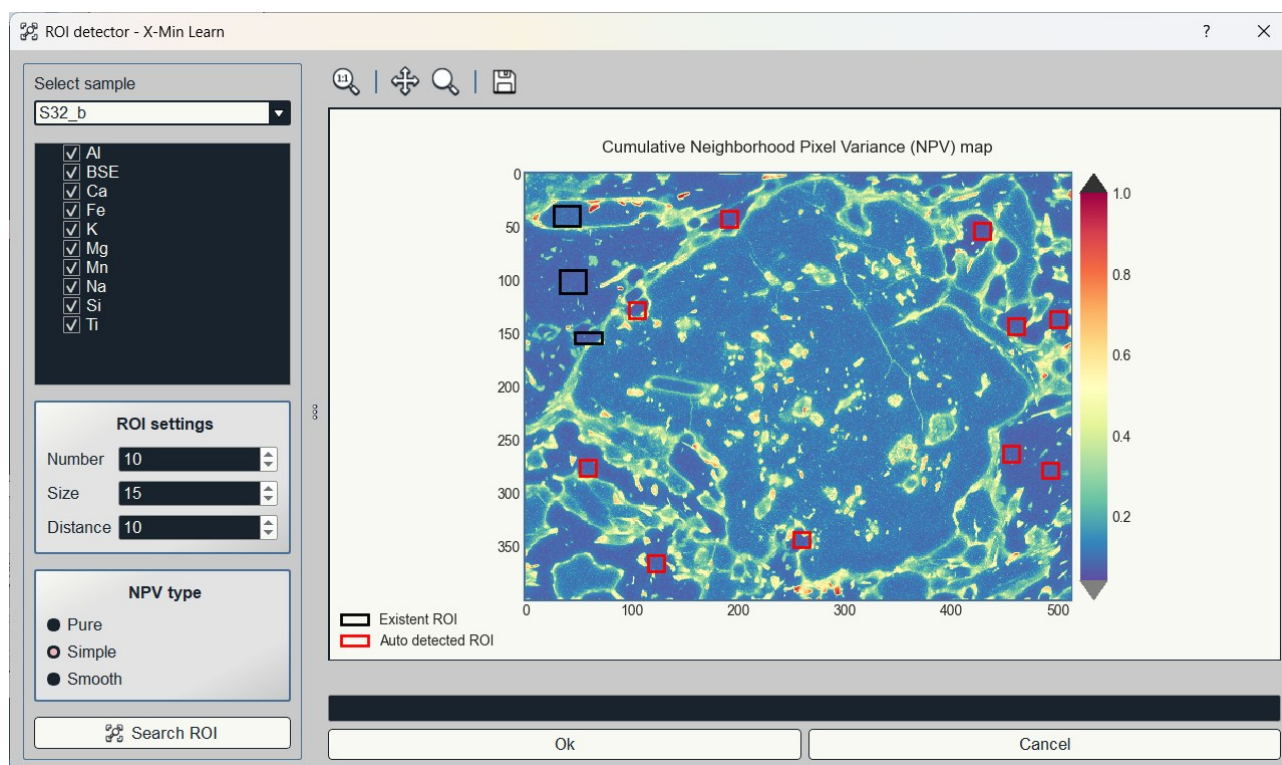


Figure 3 – ROI detector tool automatically detecting Regions of Interest based on Neighborhood Pixel Variance (NPV) maps.

Use the ROI Detector tool to automatically trace ROIs based on the analysis of Neighborhood Pixel Variance (NPV) maps, which highlight the most pixel-homogeneous areas in your sample across the selected input maps (Figure 3). Several parameters can be tweaked to customize the computation of NPVs and the ROIs tracing. Open the dialog using the dedicated button [🔍] in the toolbar.

Probability Map

Probability maps are embedded in every mineral map generated during mineral classification routines (see Mineral Classifier) by both supervised and unsupervised algorithms. They quantify the degree of classification confidence of the algorithm for each pixel of a mineral map. Probability maps



are visible in the Probability Map pane when the associated mineral map is displayed in the Data Viewer (Figure 4).

Simple image navigation options are available from the dedicated toolbar (zoom, pan, save image). Zoom and pan actions are synchronized: any zoom/pan applied to probability maps are mirrored to mineral maps and vice versa.

Probability ranges can be customized through the dedicated spin-boxes, after toggling *map scaling* [🔍]. This allows highlighting pixels with specific classification confidence.

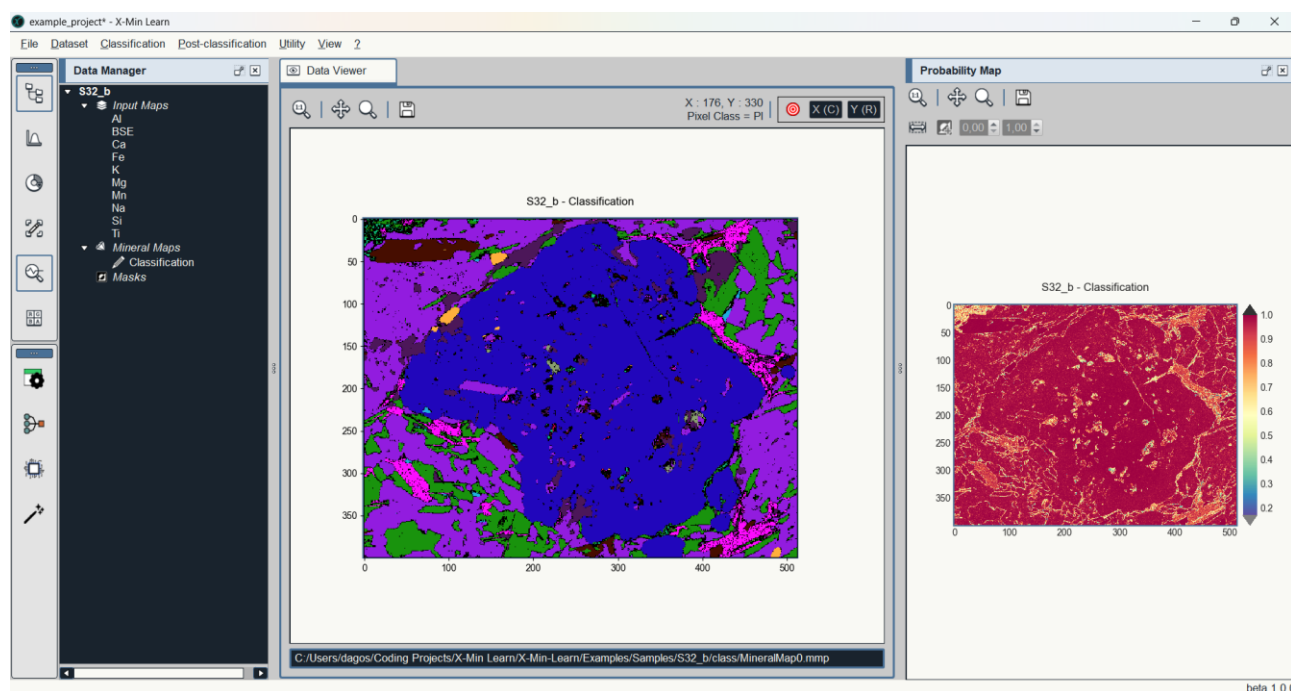


Figure 4 – The two components of a mineral map: classification map, displayed in the Data Viewer, and the linked probability map.

RGBA Map

Up to four input maps from the same sample can be combined to generate a single RGBA map. This is a well-known strategy for image analysis, widely used to visualize several single-band images simultaneously. The RGB channels allow colorizing grayscale maps, respectively using red, green and blue tints, while the color intensity reflecting the value of the pixels in each channel. The A component (alpha) adjusts transparency of the composite image, making regions with higher pixel values less transparent.

Navigation actions performed on RGBA maps (i.e., zoom and pan) are synchronized with maps displayed in the Data Viewer and vice versa.

ANALYSIS TOOLS

Analysis tools are the backbone of X-Min Learn, allowing to perform automatic mineral classifications from input maps, post-processing refinements of mineral maps and user-friendly development of custom machine learning models. The four main analysis tools, that can be opened from the *menu bar* or from the *tools toolbar*, are:



- [🔧] **Mineral Classifier**, to generate mineral maps from input data using supervised and unsupervised machine learning algorithms.
- [🔪] **Phase Refiner**, to perform post-processing refinement on classified mineral maps with basic and advanced image analysis algorithms.
- [⚙️] **Dataset Builder**, to compile ground truth dataset from your own data, which are required to build custom machine learning models.
- [🧑🏫] **Model Learner**, to train custom machine learning models from scratch in an interactive, no-code environment.

Mineral Classifier

The Mineral Classifier tool allows for full- or semi-automatic mineral classification. The tool's UI consists of three main elements (see Figure 5):

- *Data panel*, where input map data and output classified mineral maps are listed.
- *Classifier panel*, where a classification algorithm can be selected.
- *Viewer panel*, where input and output maps are displayed.

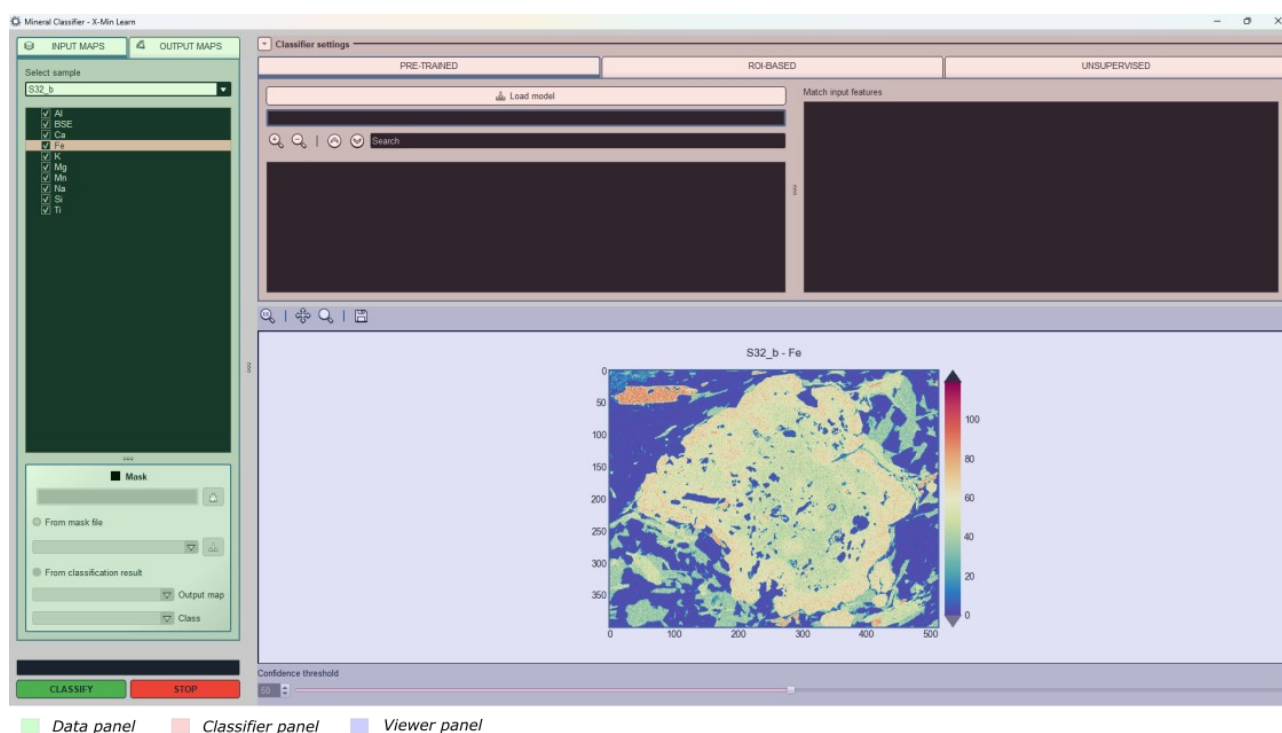


Figure 5 – Mineral Classifier tool.

In order to classify a new sample, it must be selected from the dedicated combo-box at the top of the *Data panel*, in the *input maps* tab. A sample will be listed in the combo-box only if it was previously imported in the project (see [Importing Data](#)). Once a sample is selected, the input maps data will be listed in the box below (see Figure 5). Input maps can be toggled on/off using the dedicated checkboxes, allowing full control over which data should be processed by the classification algorithms.



Three types of classifiers are available and are organized in three different tabs in the *Classifier* panel (Figure 6):

- *Pre-trained*, for custom user-trained machine learning models.
- *ROI-based*, for instance-based classifiers trained on user-drawn Regions of Interest (ROIs).
- *Unsupervised*, for clustering algorithms.

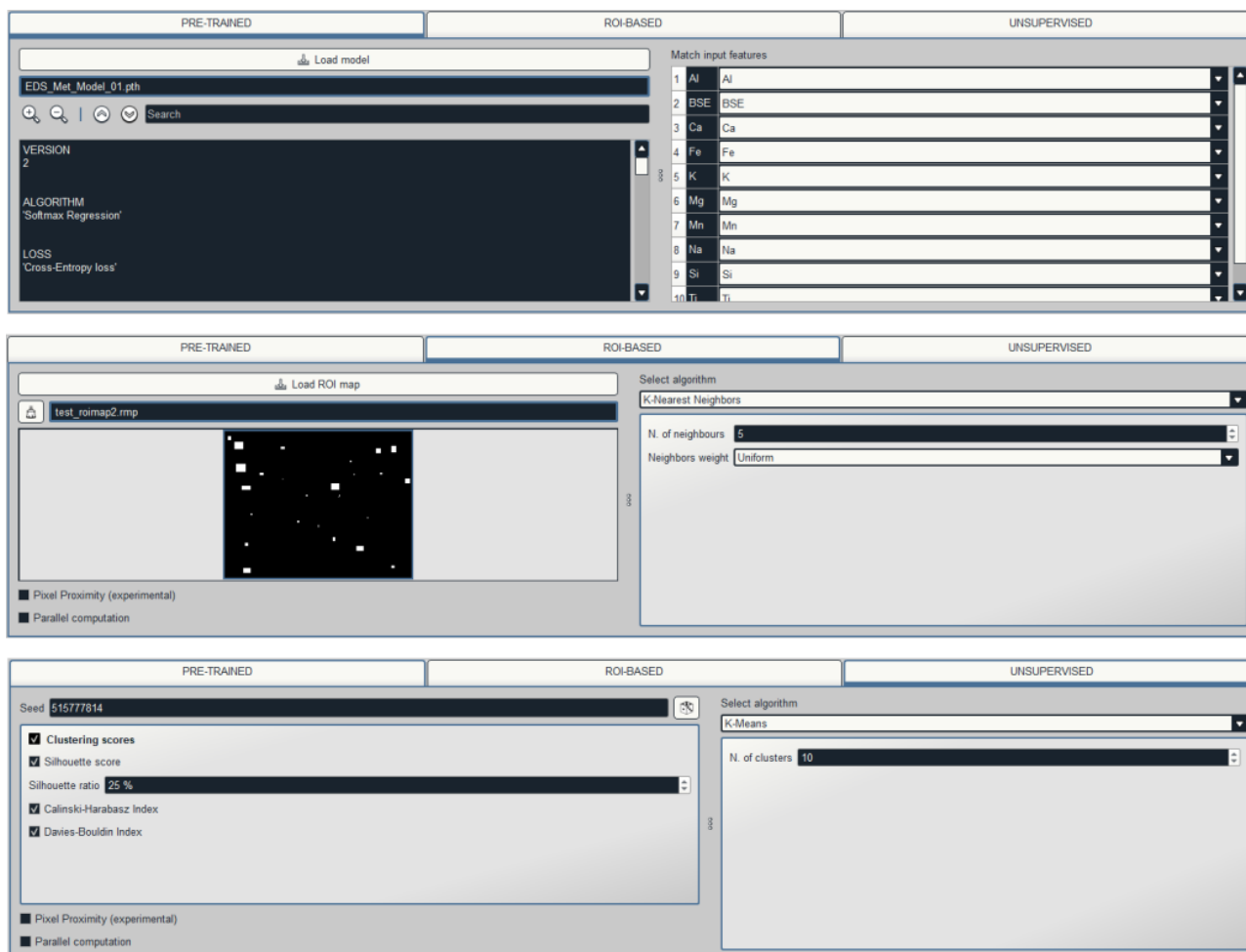


Figure 6 – Dedicated settings for the three classifier types, from top to bottom: Pre-trained models, ROI-based classifiers, Unsupervised classifiers.

Pre-trained classifiers are the quickest to set up: you are only required to load a previously trained machine learning model, by clicking the *Load model* button, and to make sure that the required model's input features match with the available input maps. This can be done interactively through the dedicated combo-boxes on the right side of the panel.

ROI-based classifiers require loading valid ROIs through the *Load ROI map* button. ROI maps can be created and managed in the [ROI Editor](#) pane. On the right side of the panel, the preferred algorithm and its parameters can be selected.

Unsupervised classifiers do not require any extra file to be imported. Instead, on the left side of the panel a random seed can be set for reproducibility purposes, and, optionally, clustering scores can be enabled for a statistical evaluation of the classification result (see Figure 7). The preferred algorithm and its parameters can be selected from the right side of the panel.



The *Viewer* panel displays the currently selected map, whether it is an input map or a mineral map. In the latter case, the confidence slider at the bottom of the panel is enabled, allowing to set a confidence threshold for the classified mineral map. Pixel values with a probability score (see [Probability Map](#)) below the threshold value will be dynamically reclassified as no data (class “_ND_”).

Masks can be included in the classification routine to focus the mineral identification on specific sample areas. Masks can be enabled from the *input maps* tab in the *Data* panel (see Figure 5), and can either be loaded from file if previously extracted (see [Extracting Masks](#)), or be generated on the fly by selecting a mineral phase from a classified map. This last approach is especially useful to identify mineral sub-phases or to highlight mineral zonation patterns (see Figure 7).

Classified mineral maps are all listed in a box at the top of the *output maps* tab in the *Data* panel (see Figure 7). To save a mineral map you must select it and click the save button next to the box.

Warning!

Unsaved mineral maps will be lost if the Mineral Classifier tool is closed.

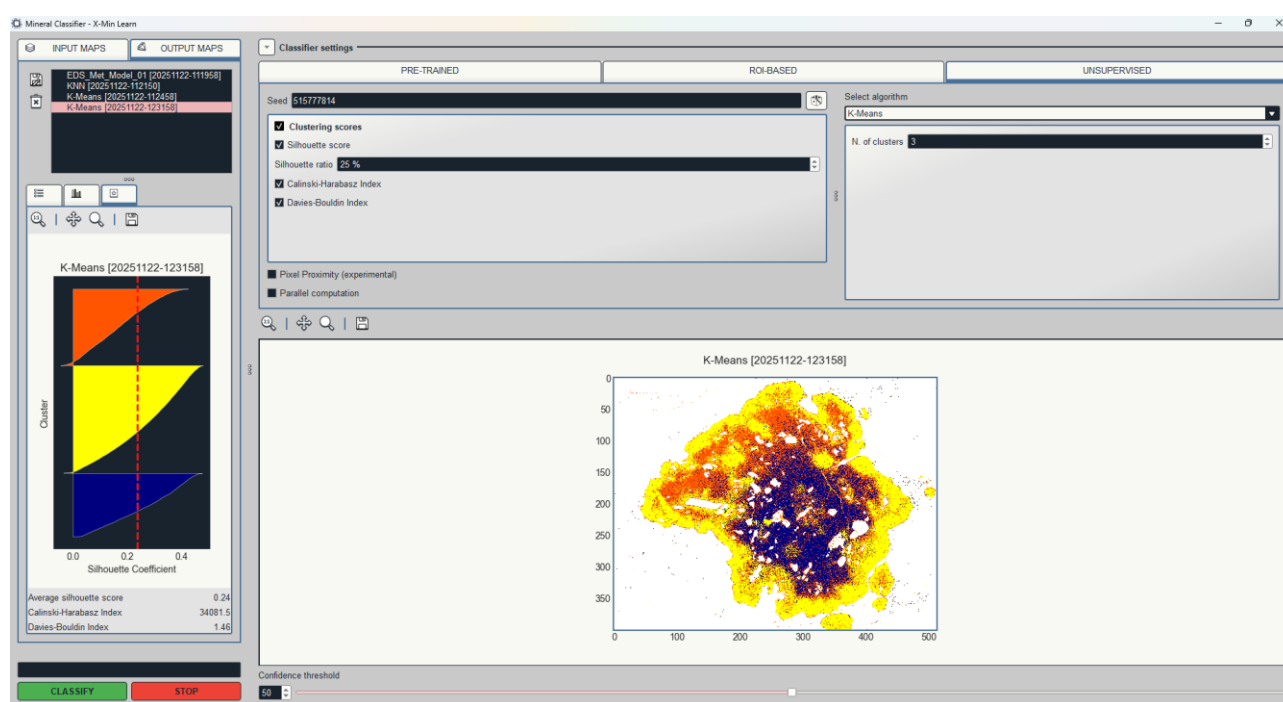


Figure 7 – Subphase analysis achieved through the application of a clustering algorithm (k-means) on masked input data. Here the mask drove the classification only on pixels assigned to a specific mineral class (garnet), identified in a previous classification routine. On the left, clustering scores are shown.

Phase Refiner

The Phase Refiner tool allows post-processing refinement of classified mineral maps. It comes with two different modalities:

- *Basic* mode (Figure 8), which allows a bulk refinement of the mineral map using a maximum frequency (or mode) filter.
- *Advanced* mode (Figure 9), which provides morphological image processing algorithms for a class-wise refinement of the mineral map.



In order to refine a classified mineral map, you must first import a sample from the dedicated combo-box in the *Map Selection* panel, on the top-left side of the window (see Figure 8). Only samples that are already loaded in the project will be listed in the combo-box (see [Importing Data](#)). After selecting a sample, its mineral maps will be listed in the box below. Only one mineral map can be refined at time; selecting a different map will cause all unsaved refinements of the current map to be lost.

The intensity of the refinement filter can be tweaked from the *Kernel* panel in both *basic* and *advanced* mode. Other options are available under the *Settings* panel: in *basic* mode, here you can fine-tune the spreading of NaN data, while in *advanced* mode you can select from six different morphological image processing algorithms to either increase or decrease the number of pixels for a specific mineral phase.

The *Legend* panel shows the percentage amount of each mineral phase, recalculated after each refinement operation. When in *advanced* mode, here you can select the class to be processed.

Click on the green “APPLY” button at the left-bottom side of the window (see Figure 8) to apply the chosen filter. There is no limit to the amount of filtering operations you can stack on the same map. You can switch between basic and advanced mode at any time to fully customize your post-processing operations. To restore the original mineral map, click the red ‘CANCEL’ button; to save the refined map click on the ‘SAVE’ button.

Warning!

Unsaved refined maps will be lost if the Phase Refiner tool is closed.

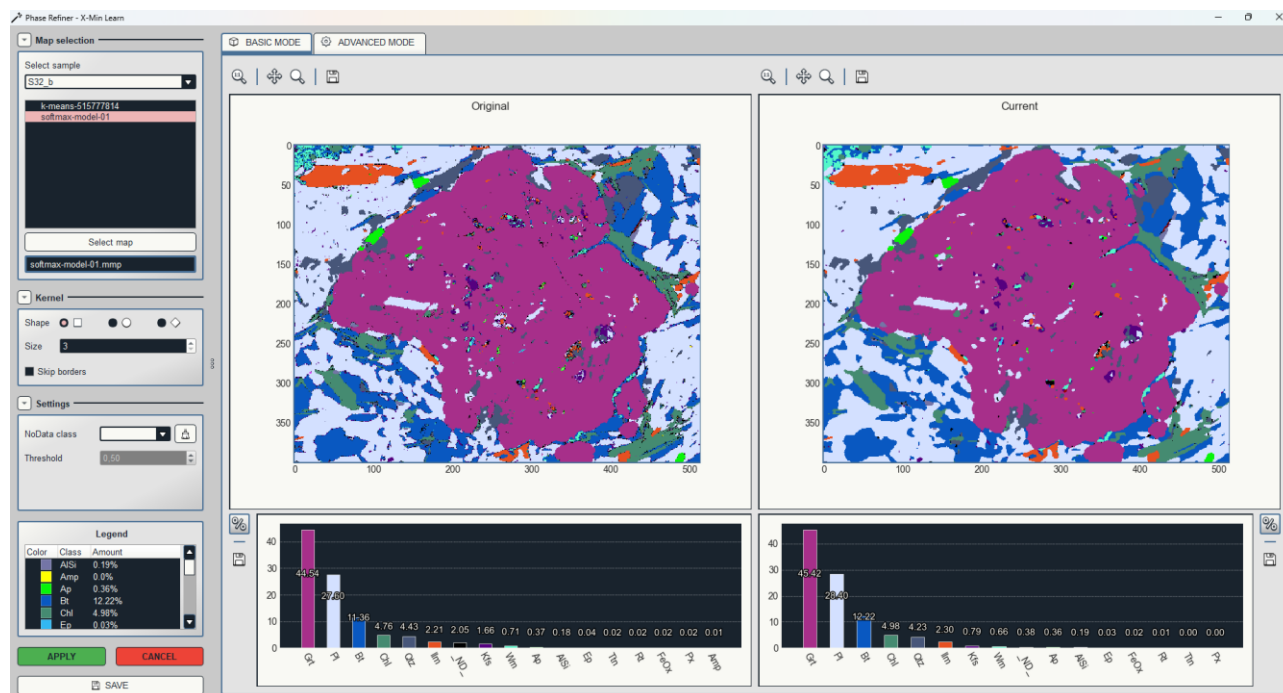
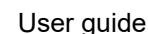


Figure 8 – Phase Refiner tool displaying the application of a mode filter in basic mode.

The *advanced* mode provides extra functionality than the basic mode. From the *Settings* panel, for example, you can specify whether pixels removed from the current mineral class should be reclassified as *NoData* (“_ND_” class) or as the nearest mineral class. The *Refined Preview* (see Figure 9) provides more detailed information on the number of pixels that will be added to or removed



Finally, the *advanced* mode allows you to draw a ROI to force the application of the filter only to a specific region of the sample, ensuring a fully customizable post-processing experience.



The Developer's Toolkit is a collection of functionalities that allow you to build a new custom machine learning model from scratch, training it on your selected mineral data. These functionalities have been condensed in two main tools:

-  Dataset Builder tool, which provides a simple and automatic way to extract numerical data from your samples and organize it into human-readable and machine-friendly datasets.
-  Model Learner tool, that allows you to train custom machine learning models from the aforementioned datasets, providing graphics and statistics to evaluate its performance.

The resulting no-code working environment ensures accessibility to the complex or technical steps required for training machine learning models, allowing non-programmers to build models that are tailored for their research needs.

The Dataset Builder tool (Figure 10) automates the organization of ground truth data in standardized, human-readable datasets. Data is seamlessly extracted, pixel by pixel, from your already classified rock samples.

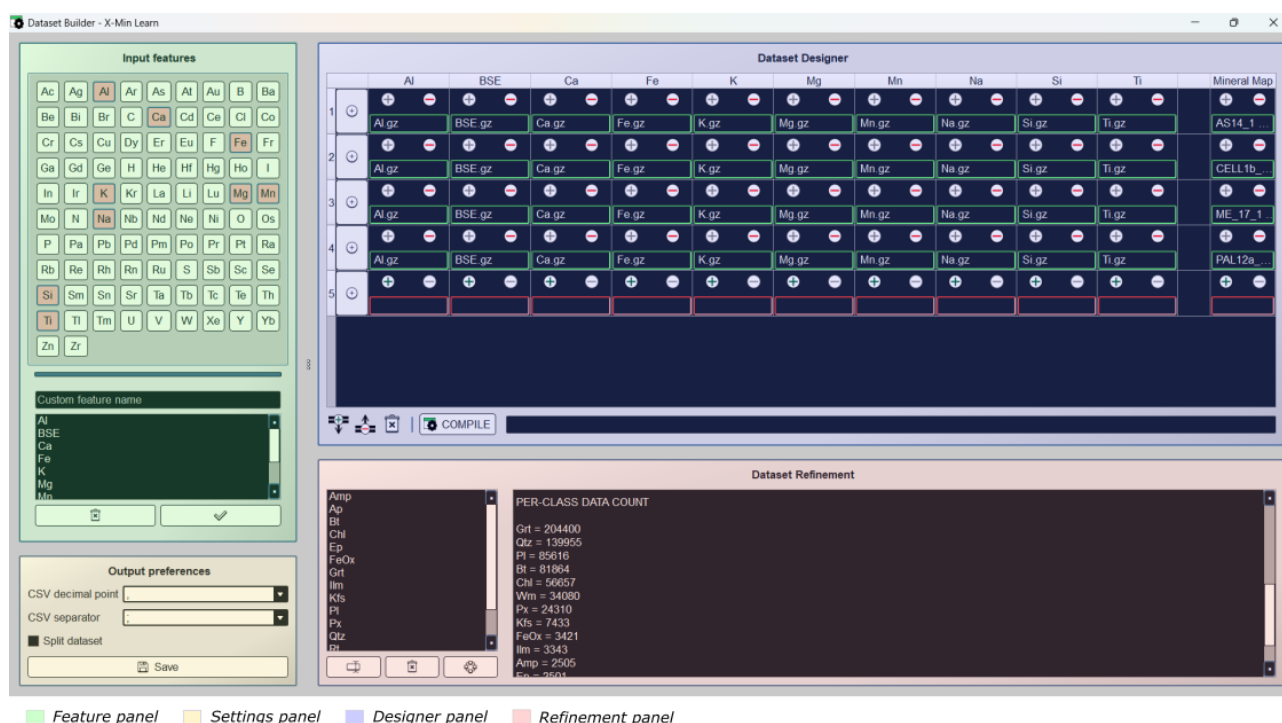


Figure 10 – Dataset Builder tool extracting ground truth data from a selection of rock samples.

From the *Input Feature* panel, you must define which input X-ray maps you are providing for each sample, either by selecting them from the list of chemical elements, or by typing in the entry below. This last option is useful to include non-elemental maps, like Back Scattered Electrons (BSE) maps. When all features have been selected, click on the *validate* button [✓] to refresh the *Designer* panel.

The *Designer* panel mimics a spreadsheet file (see Figure 10) where each row contains a single rock sample and each column indicates a specific map (X-ray maps or classified mineral map). Every cell, therefore, is meant to contain a single map that you must load from your files. The dedicated buttons allow you to add a new row [↓], remove the last row [↖] or delete the selected rows [✕]. Click on 'COMPILE' [⚙] when all your data has been loaded. This will start the automatic process of extracting pixels data from your samples to populate a new ground truth dataset.

You can refine the compiled dataset from the *Refinement* panel; namely, you can remove, rename or merge specific mineral classes. This is a crucial step, because your choice here will influence the behavior of your future models. For example, if you merge “plagioclase” and “K-feldspar” classes together in a new class named “feldspar”, future models trained using this dataset will always classify both minerals as “feldspars”. Finally, the dataset can be saved from the *Settings* panel, where you can select the decimal and separator characters of the output csv file.

Warning!

Unsaved datasets will be lost if the Dataset Builder tool is closed.

Model Learner

The Model Learner tool takes as input a ground truth dataset file compiled using the Dataset Builder tool and processes it to extract the “knowledge” contained therein. The tool ensures that the essential



steps required to train a new machine learning model are followed in order as you progress through a guided routine. The interface consists of two main blocks (see Figure 11):

- The *Settings panel*, on the left, where the main parameters that control learning sessions can be tweaked.
- The *Training panel*, on the right, where all the steps required for training a new machine learning model are organized sequentially.

The *Settings* panel is further subdivided in several sub-panels (Figure 11), from top to bottom:

- *Random Seed Generator*, that allows you to set a random seed for reproducibility purposes.
- *Ground Truth Dataset*, where you must load a dataset, required for training a new model.
- *Update Existent Model*, where you can optionally load a previously trained model in order to update it using new training data, rather than creating a completely new model.
- *Hyperparameters*, allowing you to tweak hyperparameters such as learning rate, weight decay, momentum, number of epochs (iterations) and batch size.
- *Learning Preferences*, where you can select the algorithm, the optimizer and other optional settings like polynomial feature mapping, which generates artificial features based on the available ones, potentially increasing the efficiency of the model at the cost of higher chances of overfitting.

The *Training* panel contains 4 sections which reflect the steps required for training a model:

- *Split Dataset*, to divide the ground truth dataset into train, validation and test sets. Choose the ratio of each set using the dedicated spin-boxes. After the green 'SPLIT' button is clicked, three interactive lists, on the right, allow you to inquire the number of examples in each set.
- *Balance Train Set*, to optionally apply balancing algorithms on the train set if imbalanced data occurs. This section should be skipped if the train set has comparable amounts of examples for each mineral class. The *balancing table* allows you to increase and/or decrease the number of examples in every class, by changing the values in the "After balancing" column. To actually add synthetic data and remove redundant one, you are required to select an over-sampling and/or an under-sampling algorithm from the dedicated combo-boxes (check *imbalanced-learn* user-guide¹ for algorithms info). The green 'Start' button initiates the balancing operation, and multiple operations can be stacked sequentially. The 'Cancel' button lets you discard all operations, restoring the original train set data.
- *Learning Progression*, to monitor learning sessions through interactive graphics. This section represents the most time-consuming step, where hyperparameters must be fine-tuned to optimize the model performance. To launch a new learning session, click the green 'LEARN' button on the bottom-left side of the window (see Figure 11). The model is trained on the train set and then tested on the validation set, in order to evaluate its effectiveness on data that remains hidden from the machine. The learning session can be monitored by inspecting loss and accuracy curves which update live during training operations. They are especially useful to understand if the model is converging correctly and to highlight potential overfitting.

¹ https://imbalanced-learn.org/stable/user_guide.html#user-guide



Settings panel Training panel

Figure 11 – Model Learner tool displaying an active learning session. Screenshot taken from a vertical screen to fit the full layout.



- *Model Evaluation*, to statistically evaluate the performance of the model on train, validation and test sets. This section contains confusion matrices and f1 scores that are computed at the end of every learning session. They enable a class-by-class evaluation of the model's accuracy; in particular, confusion matrices allow you to inspect which classes are most frequently confused with one another.

Usually, learning sessions are launched multiple times with different hyperparameter settings until satisfactory performance is achieved. When you are satisfied with your model's performance you can finally test it one last time on the test set, by clicking on the 'TEST MODEL' button, at the bottom left of the window. You are encouraged to NOT train your model any further at this point, because this final assessment of its performance is meant to simulate the behavior of your model with new unseen data. If you keep training it in order to increase this final score you are effectively defying the purpose of the test set. Models can be saved for future uses by clicking the 'SAVE MODEL' button.

Warning!

Unsaved models will be lost if the Model Learner tool is closed.

EXTRACTING MASKS

Masks are powerful objects that allow you to visually clip input and mineral maps without changing their data. They are especially useful during mineral classification routines to focus the analysis only on specific areas of the sample, enabling, for example, mineral subphase identification (see Figure 7). You can also employ them to better highlight pixel variations in your input data, as illustrated in Figure 12.

Masks can be extracted from different sources; the *Extract Mask* button , available in most Panes, indicates that you can generate a mask from that specific data source. In particular, you can generate masks in:

- The Histogram pane, from a range of values highlighted by the *scaler* (see Figure 2). The *Extract Mask* button is located in the toolbar, next to the *scaler* button .
- The Mode pane, from a mineral class listed in the *legend*. Right-click the class to access the *Extract Mask* option.
- The ROI Editor pane, from a user-drawn ROI. The *Extract Mask* button is located in the toolbar, next to the *Add ROI* button .
- The Probability Map pane, from a range of confidence scores highlighted by the *scaler*. The *Extract Mask* button is located in the toolbar, next to the *scaler* button .

Masks must be loaded in the *Masks* data type entry of a specific sample (see Importing Data), and can be toggled on/off by clicking their checkbox (see Figure 12). Active masks are applied only to the sample they belong to. More than one mask can be toggled at the same time: in such scenario X-Min Learn automatically generates a composite mask on the fly, based on the mask merging rule (see Preferences).

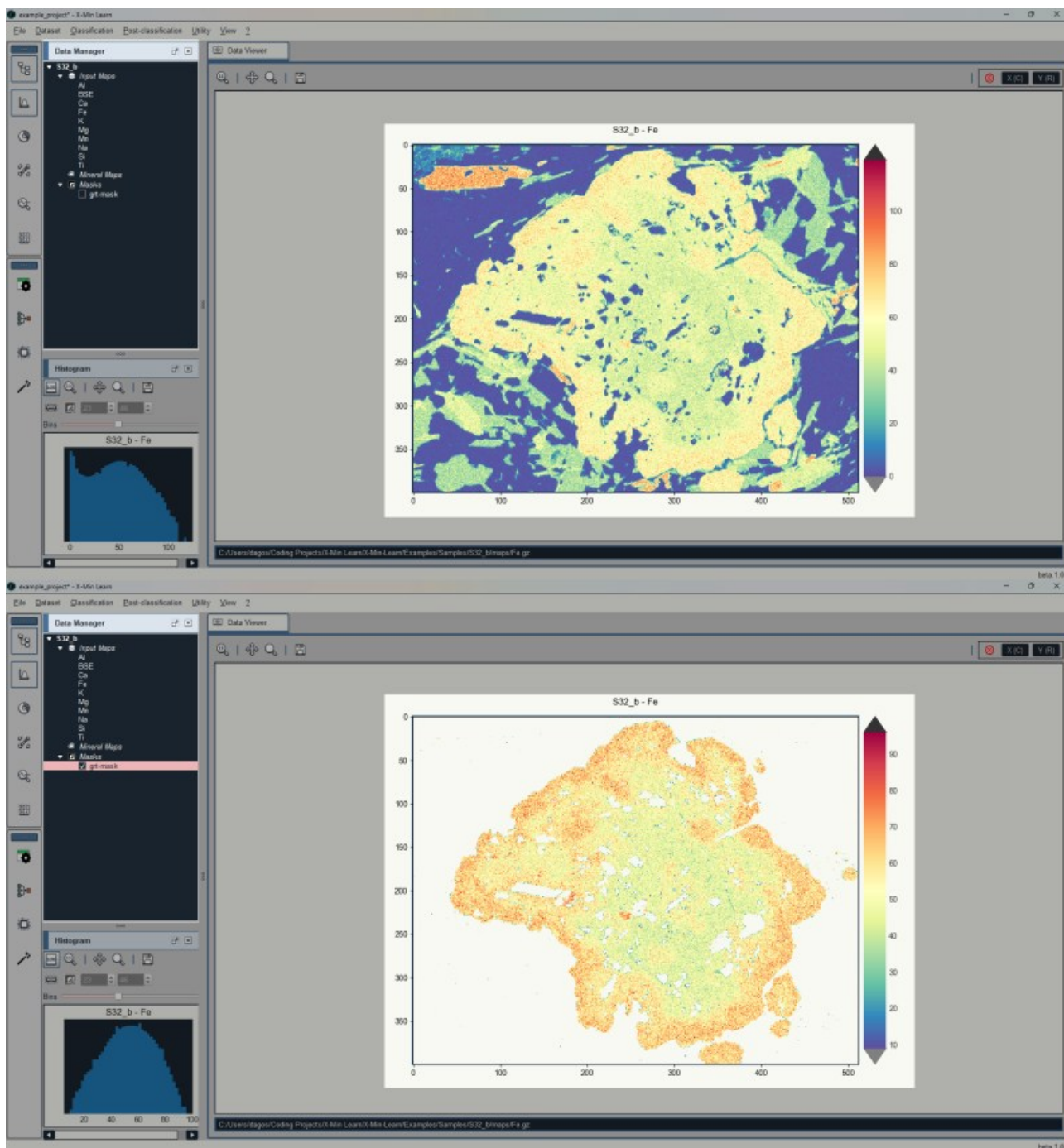


Figure 12 – Mask applied from the Data Manager to visually clip pixels from the "Fe" X-ray map that belong to the class "Garnet".

UTILITIES

X-Min Learn includes several QoL tools for minor yet convenient operations. These tools are not grouped in a dedicated toolbar, instead they can be accessed from the *Menu bar* (see Figure 1).

Image Conversion

Image conversion tools are accessible from the Utility menu > *Convert*, and can be employed to transform image data from common formats (*bmp*, *jpg*, *png*, *tiff*) to X-Min Learn-compatible map data formats (*txt*, *gz*, *mmp*). Image conversion tools consist of:



- *Image To Input Map* tool (Figure 13), to convert images to input map data (*txt* or *gz* formats). Images can be loaded with the *Import* button []; the tool shows a preview of loaded images, which can be converted by clicking the green 'Convert' button. This tool expects you to convert only single-band images; if, for any reason, you need to convert multi-band images (e.g., RGB images), make sure that the 'Split multi-channel images' option is checked. This will convert each channel of your multi-band images into separate input maps.

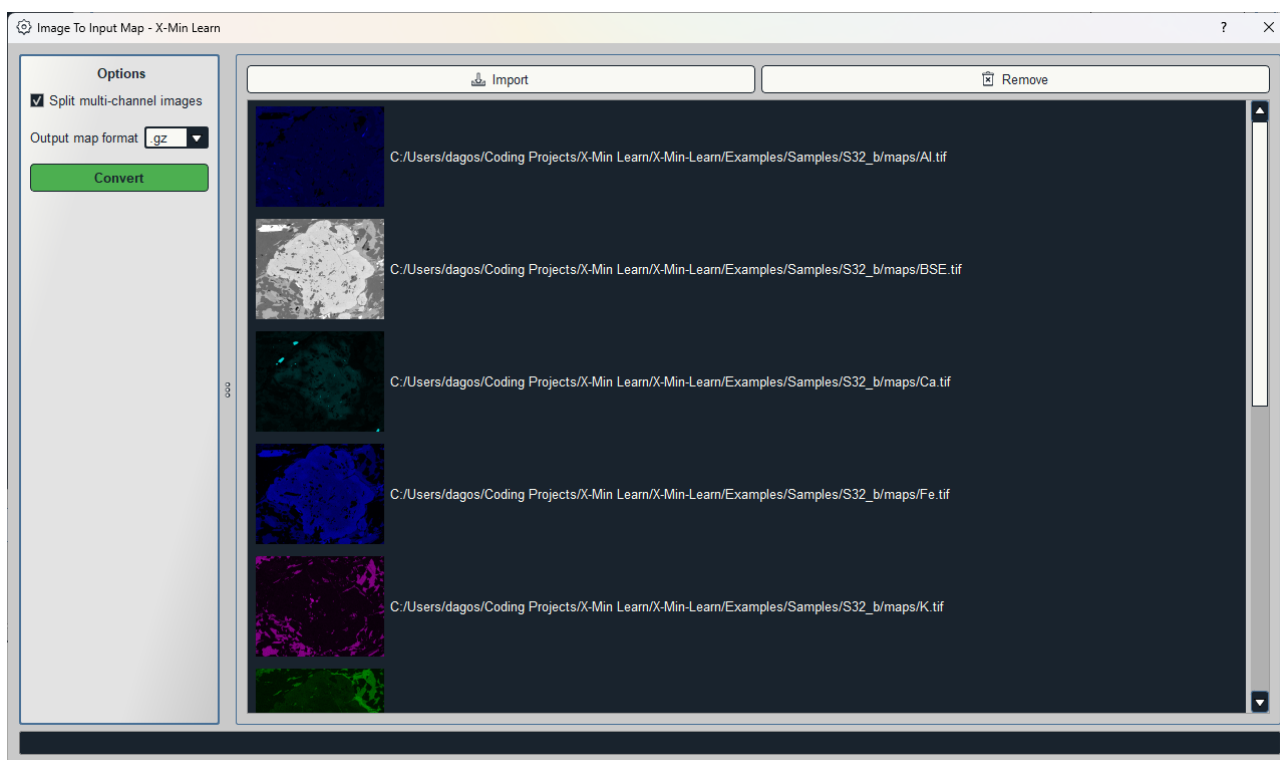


Figure 13 - X-ray maps data in TIF format loaded into the *Image To Input Map* tool for conversion.

- *Image To Mineral Map* tool (Figure 14), to convert images to classified mineral map data (*mmp* format). Import your image through the *Import* button [], and convert it to a mineral map by clicking the green 'Convert' button. The tool will automatically segment the image in classes, which are listed in the dedicated legend. The *Color delta* parameter can be tweaked to identify more or less classes from the original image, with lower values determining a higher number of classes detected (i.e., a more fine-grained color detection). Here over-segmentation is preferable to under-segmentation, because over-segmented classes can be merged together once the map is loaded in the Data Manager (see Mode pane). When you are satisfied with the segmentation result, click on the 'Save' button to save the converted mineral map.

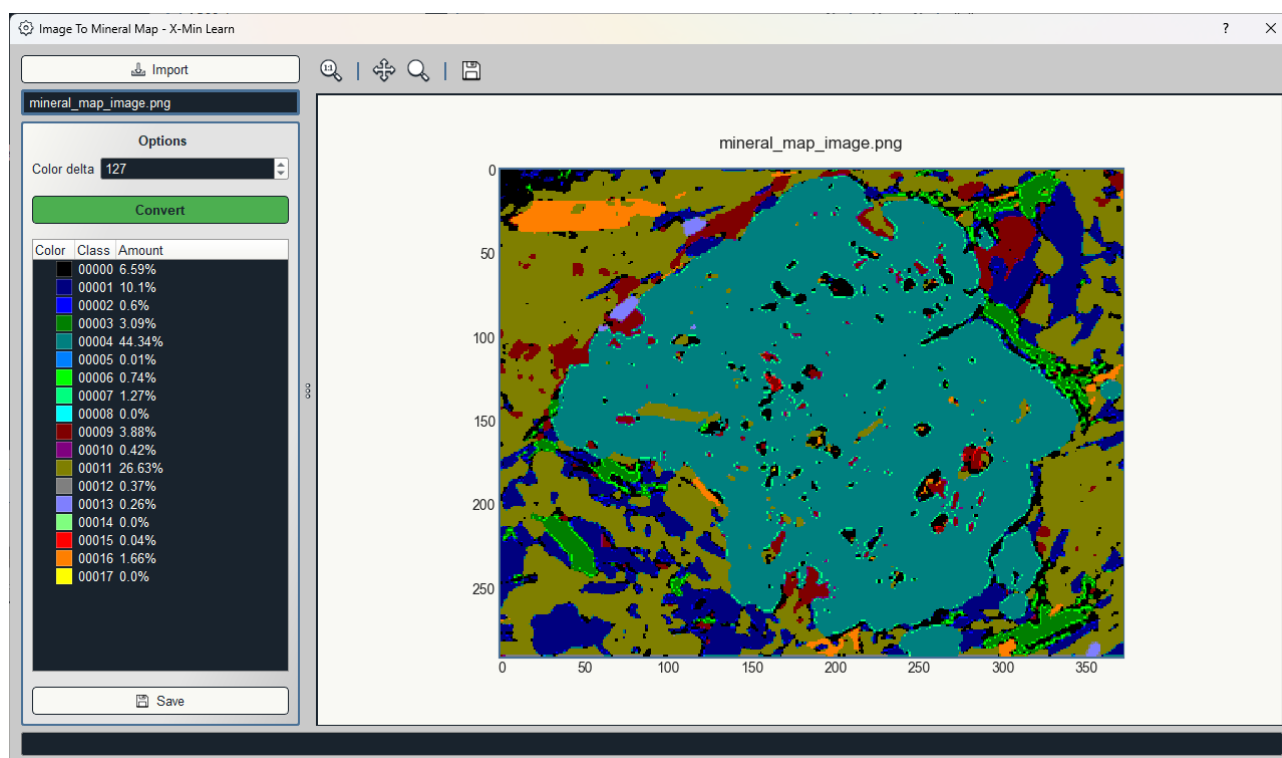


Figure 14 – PNG image segmented in mineral classes using the Image to Mineral Map tool.

Dataset Manipulation

Dataset Manipulations tools (Figure 15) are accessible from the Dataset menu and allow managing the ground truth datasets generated with the [Dataset Builder](#) tool. In particular:

- The *Sub-sample Dataset* tool allows you to extract a portion of data from an existing dataset and use it to generate a new one. Once the original dataset is imported, you can use the class selection widget to drag & drop the mineral classes you want to include in the new dataset. After clicking the green 'Sub-sample' button, a new dataset will be generated and populated only with the ground truth data associated with the selected classes.
- The *Merge Datasets* tool allows you to merge two or more datasets into one. This tool is helpful to extend the amount of training data you want to feed in your custom models with, since the [Model Learner](#) tool only accepts one single ground truth dataset as input. After all datasets have been loaded, click on the green 'Merge' button to unify their data into a new merged dataset. Be aware that only datasets that share the exact same input features can be merged together. Make sure that is the case by inspecting the first rows of each of the imported datasets through the dedicated *Preview* panel (see [Figure 15](#)).

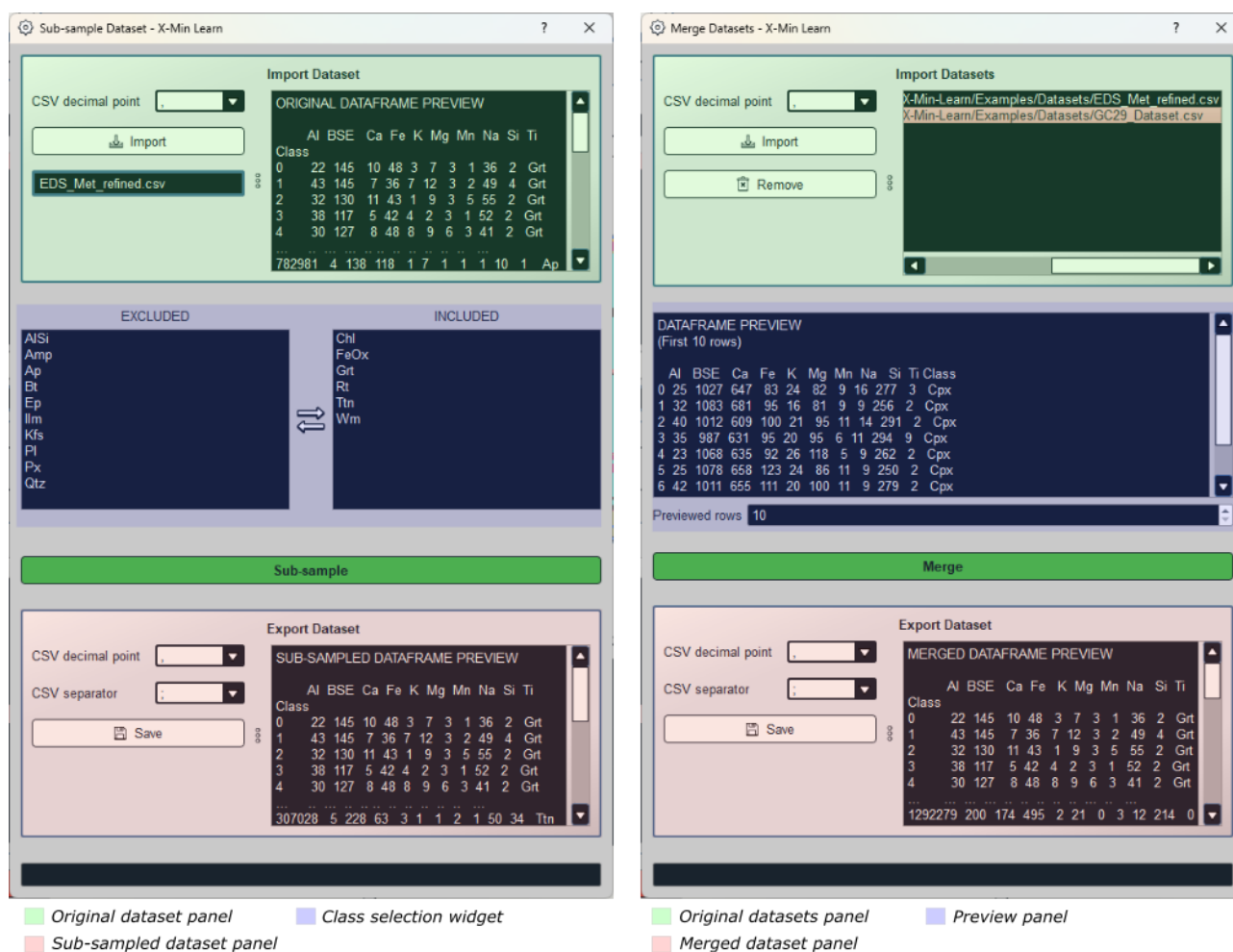


Figure 15 – (Left) Creation of a new ground truth dataset using the Sub-sample Dataset tool. The new dataset only includes "Chl", "FeOx", "Grt", "Rt", "Ttn" and "Wm" mineral classes. (Right) Creation of a new ground truth dataset by combining data from two existing datasets using the Merge Datasets tool.

Dummy Maps

One downside of using pre-trained machine learning models is that they always require the exact same type of input maps they were trained with. Sometimes you may need to classify a rocks sample but you are missing one specific X-ray map required by your favorite model. The *Dummy Maps Builder* tool (Figure 16) allows generating artificial noisy maps with near-zero values, randomized through a Gamma distribution function. These maps can be used as a placeholder for missing maps in a pre-trained model, BUT ONLY if you are certain that such specific map, if collected, would have just produced noisy pixels (see example in Figure 16).

From the *Options* panel you can define the map size (width and height in pixels) and tweak the function parameters for different pixel distributions. Click the green 'RANDOMIZE' button to generate a new dummy map and check its resulting histogram in the viewer. Once you are satisfied, you can save the dummy map as a new input map by clicking the 'Save' button.

Warning!

Always prefer collecting real data rather than generating synthetic one. Use this function only if you cannot use a different model or collect the missing map.

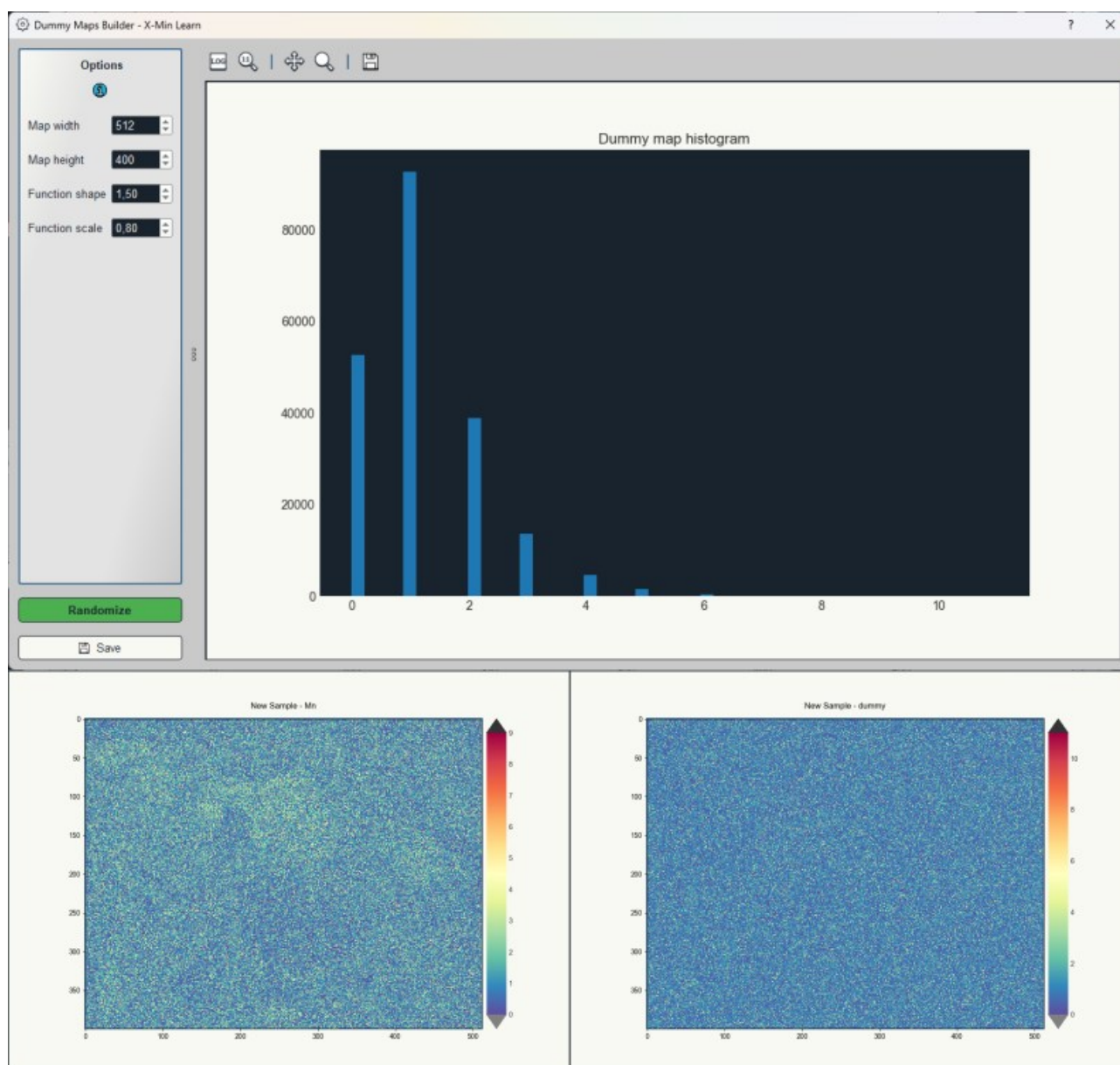


Figure 16 – Dummy map (bottom-right) generated with the Dummy Maps Builder tool (top), compared with a real noisy Manganese X-ray map (bottom-left).

PREFERENCES

The Preferences dialog allows you to customize certain aspects of X-Min Learn. You can access it from the File menu. Preferences are organized in three separate tabs (Figure 17):

- **Interface** tab, which allows you to control certain elements of the GUI, namely:
 - Font size (defaults to 10).
 - DPI Scaling (defaults to False). Enable this if you are using high DPI screens.
 - Tabbed tools (defaults to True). If enabled, opened Analysis Tools will be docked automatically in the central widget of the main window (see Figure 1).
 - Smooth resize (defaults to True). If enabled, you will get smooth feedback when resizing widgets. Disable this option if you experience lag when resizing widgets.
- **Plot Style** tab, that allows you to customize the look of rendered data. The only available settings are under the *ROI* section, and include:



- Color (default is #19232d), which determines the color of ROIs outline and filling.
- Selection color (default is #55ffff), which controls the ROIs color when selected.
- Filled (defaults to False), which determines if ROIs should have a fill color.
- **Data** tab, that includes options affecting various data types, namely:
 - Decimal precision (defaults to 2), that controls the number of displayed decimal digits.
 - Mineral phase count warning (defaults to 25). When importing a mineral map, a warning message will show up if the number of mineral classes exceed this threshold.
 - Mask merging rule (defaults to 'Union'). This controls how the application should merge two or more masks when generating a composite mask on the fly (see [Extracting Masks](#)). If 'Intersection' is selected, pixels that are visible through at least one mask will be visible through the composite mask. If 'Union' is selected, only pixels that are visible through all masks will be visible through the composite mask.
 - Extended log models (default to False). Enable this option to include advanced information in your models' log files. This is mostly required for debugging operations.

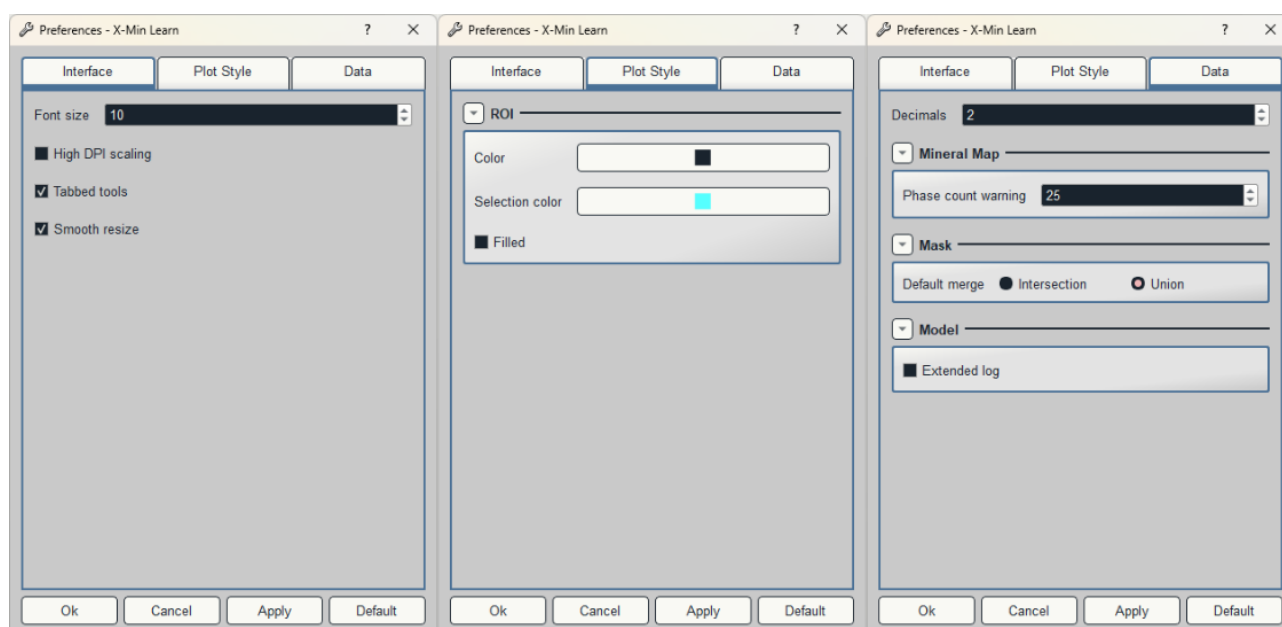


Figure 17 – All the options available in the Preference dialog.